Cultural Homophily in Tennis Doubles

Grand Slam Analysis 2018–2025

Prepared for supervisor review · May 2026 · Code: `code/analysis/homophily_analysis.ipynb`

Key Findings

- **Language proximity positively predicts match wins** in the updated filtered sample; the marginal-effect table reports the probability-scale estimate.
- **No robust pressure advantage under tiebreak stress:** language proximity is not statistically significant for tiebreak outcomes in the baseline specification.
- **Homophily shares are stable pre-2024** (~40% same-nationality, ~53% same-language from 2018–23), then drop sharply in 2024–25 — an artifact of expanded draw sizes (134 vs. ~62 matches per Grand Slam).
- **Ranking and career win rate dominate** match outcomes, as expected.

1. Data Overview

- **Matches:** 2,105 retained from 2,192 raw matches | 2,042 Grand Slams | 63 Olympics
- **Retirements/walkovers:** 87 observations dropped using the score-completion rule.
- **Years:** 2018, 2019, 2021, 2022, 2023, 2024, 2025 (2020 excluded — COVID)
- **Tournaments:** Australian Open, Roland Garros, Wimbledon, US Open, Olympics (2021, 2024)
- **Analysis sample:** 3,363 team-level observations (1,681 GS matches × 2 teams)

Player rankings and biographical profiles (nationality, language, coach) are sourced from ATP Tour snapshots merged to each match. Language variables derive from the CEPII Gravity dataset: *same language (official)* uses common official language; *language proximity* uses the broader ethno-linguistic criterion.

2. Pressure Outcomes

2.1 Tiebreak Observations

A **regular tiebreak** (7-point, played at 6-6 in sets 1 or 2) occurred in **45.1% of retained matches**. A **match tiebreak** (10-point super-tiebreak, replacing a full third set) was played in 24 matches. For regression purposes both regular and match tiebreaks are combined under *any tiebreak*.

Pressure Outcome	N	Share
Set-1 tiebreak (7-pt)	521	24.8%
Set-2 tiebreak (7-pt)	564	26.8%
Set-3 regular tiebreak (7-pt)	230	10.9%
Match tiebreak / super-tb (10-pt, set 3 ≥ 8)	24	1.1%
Any tiebreak (all types)	1,058	50.3%
Match went to 3 sets	895	42.5%
Comeback win (winner lost set 1)	393	18.7%

N = 2,105 retained matches after dropping 87 likely retirements/walkovers.

2.2 By Tournament

Tournament	N Matches	Reg. Tiebreaks	TB Rate	Match Tiebreaks	Comeback Wins	Comeback Rate
Australian Open	568	261	46.0%	0	109	19.2%
Olympics	63	28	44.4%	20	13	20.6%
Roland Garros	567	200	35.3%	0	114	20.1%
US Open	330	154	46.7%	0	55	16.7%
Wimbledon	577	307	53.2%	4	102	17.7%

2.3 Comeback Wins

A comeback win is defined as the eventual match winner having lost the first set. This occurred in **393 matches (18.7%)** of the full sample, and in **43.9%** of three-set matches — meaning the team that won the first set subsequently lost the match roughly two-fifths of the time.

3. Cultural Homophily Across Olympic Cycles

The descriptive tables below show the share of pairs (averaged over both teams in each match) with each homophily characteristic, broken out separately for the Tokyo 2021 and Paris 2024 cycles. Grand Slam matches only — the Olympics tournament is excluded because same-nationality pairing is compulsory there.

3.1 Tokyo 2021 cycle

Table 2A. Tokyo 2021 cycle — Team composition across period

	Same Nationality (%)	Same Language (%)	Ling. Proximity (%)	N matches
Full Sample	**37.3**	**50.3**	**50.8**	**2042.0**
Pre-Tokyo (2018–19)	39.8	53.2	54.1	503.0
Tokyo Prep (2020–21)	41.2	54.1	54.1	182.0
Post-Tokyo (2021–22)	38.4	53.0	53.8	302.0

Tokyo panel includes 2020 as Tokyo Prep, excluding Wimbledon 2020 because Olympic doubles was not yet in the build-up path.

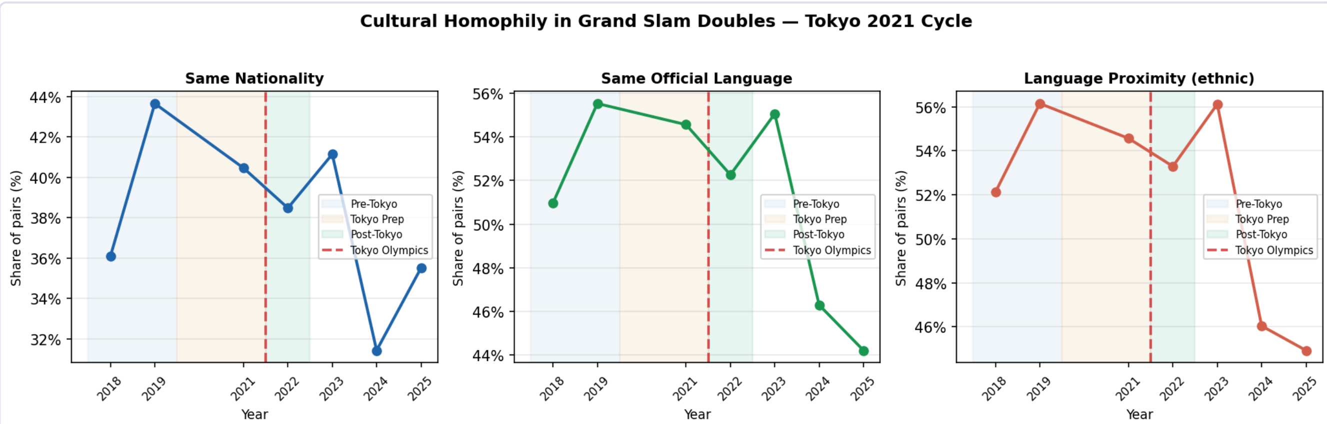

Figure 1. Tokyo 2021 cycle homophily trends. Shaded bands mark Pre-Tokyo, Tokyo Prep, and Post-Tokyo periods.

3.2 Paris 2024 cycle

Table 2B. Paris 2024 cycle — Team composition across period

	Same Nationality (%)	Same Language (%)	Ling. Proximity (%)	N matches
Full Sample	**37.3**	**50.3**	**50.8**	**2042.0**
Pre-Paris (2022)	38.5	52.3	53.3	243.0
Paris Prep (2023–24)	34.8	49.6	49.8	571.0
Post-Paris (2024–25)	35.7	44.9	45.6	484.0

Paris panel treats the 2023–24 Grand Slams as the Paris Prep window, with 2024 US Open and 2025 matches as Post-Paris.

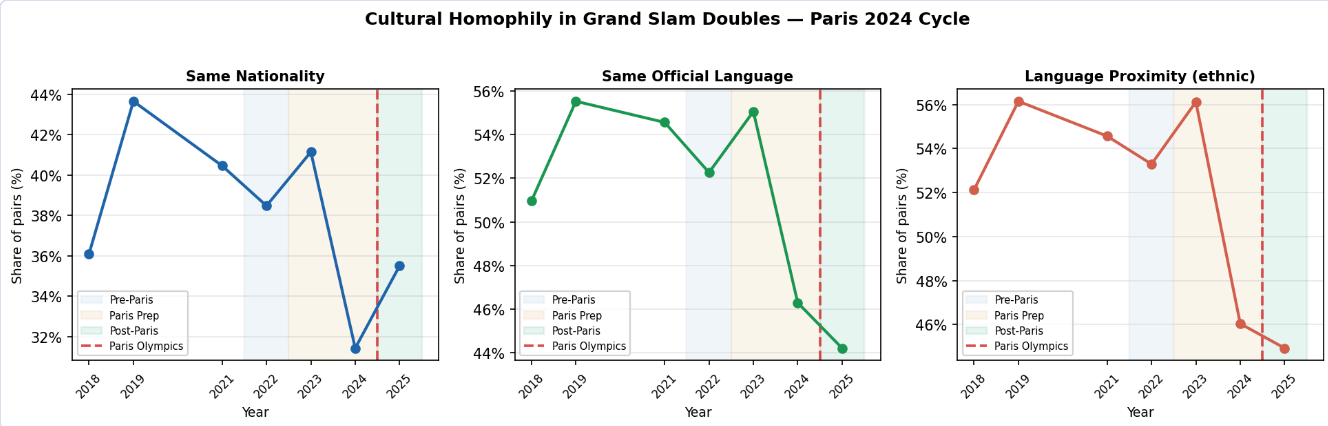

Figure 2. Paris 2024 cycle homophily trends. Shaded bands mark Pre-Paris, Paris Prep, and Post-Paris periods.

3.3 Winner vs. Loser Team Homophily

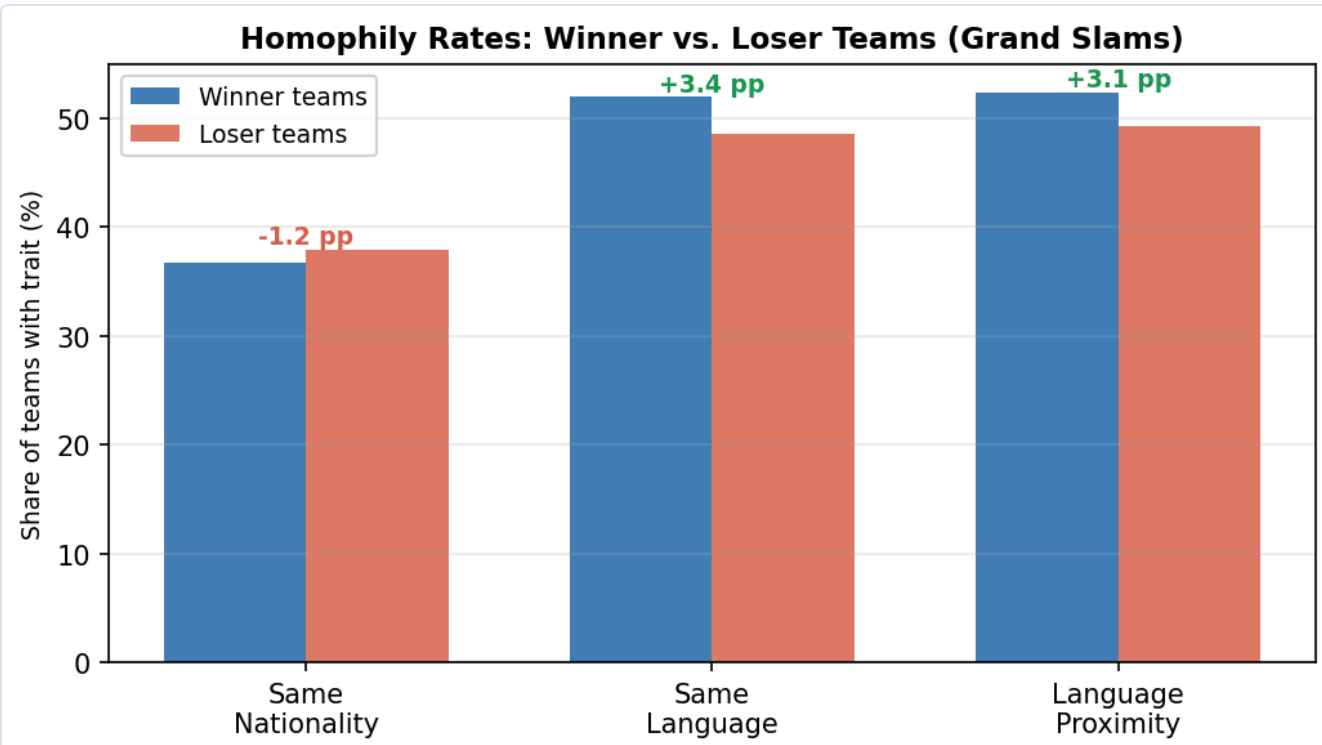

Figure 3. Percentage of teams with each homophily trait, split by match outcome. Annotation shows the winner-minus-loser gap.

Winner teams are more likely to share a language (++3.4 pp) and to be linguistically proximate (++3.1 pp) than loser teams. Same-nationality is marginally more common among losers (38.0% vs 36.7%).

4. Baseline Regressions

Specification: each match contributes two team-level observations (winner = 1, loser = 0). Regressors include language proximity (ethnic) and ranking controls (team mean ranking, opponent mean ranking, within-team rank gap, and a top-100 singles indicator). Fixed effects for tournament×year and round. Standard errors are clustered by match.

4.1 Match Win

	Logit	
Language proximity (ethnic)	+0.2579***	(0.0739)
Team average doubles ranking	-0.0059***	(0.0008)
Opponent average doubles ranking	+0.0033***	(0.0004)
Teammate rank gap	+0.0019***	(0.0006)
Top-100 singles player	+0.1565	(0.1487)
Observations		3,363
R^2 / Pseudo-R^2		0.068

Table 3. Match Win. Outcome = 1 if team won. Language proximity only. Tournament×year and round FE. SE clustered by match.

	Average marginal effect, pp	
Language proximity (ethnic)	+5.89***	(1.68)
Team average doubles ranking	-0.14***	(0.02)
Opponent average doubles ranking	+0.07***	(0.01)
Teammate rank gap	+0.04***	(0.01)
Top-100 singles player	+3.55	(3.35)
Observations		3,363

Table 3M. Match Win — average marginal effects. Effects are percentage-point changes in Pr(win).

Significance: *** $p < 0.01$ ** $p < 0.05$ * $p < 0.10$ Standard errors in parentheses.

Language proximity is associated with a 5.9 percentage-point higher probability of winning the match; this estimate is statistically significant ($p = 0.000$).

4.2 Tiebreak Win (pressure outcome)

Sample restricted to matches with any tiebreak, regular or match tiebreak (N = 861 matches). Outcome: team won at least one tiebreak.

	Logit	
Language proximity (ethnic)	+0.0733	(0.1098)
Team average doubles ranking	-0.0007	(0.0007)
Opponent average doubles ranking	+0.0010**	(0.0005)
Teammate rank gap	-0.0007	(0.0007)
Top-100 singles player	-0.1463	(0.2177)
Observations		1,722
R^2 / Pseudo-R^2		0.024

Table 4. Tiebreak Win — Logit (sample: matches with any tiebreak, 7p/10p). Outcome = 1 if team won any tiebreak. Language proximity only. Tournament×year and round FE. SE clustered by match.

	Average marginal effect, pp	
Language proximity (ethnic)	+1.64	(2.45)
Team average doubles ranking	-0.01	(0.02)
Opponent average doubles ranking	+0.02**	(0.01)
Teammate rank gap	-0.02	(0.01)
Top-100 singles player	-3.32	(5.01)
Observations		1,722

Table 4M. Tiebreak Win — average marginal effects. Effects are percentage-point changes in Pr(win any tiebreak).

Significance: *** $p < 0.01$ ** $p < 0.05$ * $p < 0.10$ Standard errors in parentheses.

Language proximity is associated with a 1.6 percentage-point higher probability of winning a tiebreak in tiebreak matches; this estimate is not statistically significant ($p = 0.504$).

4.3 Comeback Win (pressure outcome)

Sample: three-set Grand Slam matches *conditional on the team having lost set 1*. Outcome = 1 if the team wins the match (i.e., a comeback). Tournament×year fixed effects are included by encoding each tournament-year cell; cells with no within-cell variation in the outcome are dropped. Standard errors are clustered by match.

	Logit	
Language proximity (ethnic)	-0.3309*	(0.1759)
Team average doubles ranking	-0.0038**	(0.0016)
Opponent average doubles ranking	+0.0011	(0.0008)
Teammate rank gap	+0.0011	(0.0015)
Top-100 singles player	+0.3531	(0.3707)
Observations		727
R^2 / Pseudo-R^2		0.106

Table 5. Comeback Win — Logit (3-set GS matches; conditional on losing set 1). Outcome = 1 if team wins given it lost set 1. Language proximity only. Tournament×year FE (cells with no within-cell variation dropped). SE clustered by match.

	Average marginal effect, pp	
Language proximity (ethnic)	-7.04*	(3.72)
Team average doubles ranking	-0.08**	(0.03)
Opponent average doubles ranking	+0.02	(0.02)
Teammate rank gap	+0.02	(0.03)
Top-100 singles player	+7.54	(7.91)
Observations		727

Table 5M. Comeback Win — average marginal effects. Effects are percentage-point changes in Pr(win | lost set 1).

Significance: *** $p < 0.01$ ** $p < 0.05$ * $p < 0.10$ Standard errors in parentheses.

Language proximity is associated with a 7.0 percentage-point lower probability of completing the comeback; this estimate is marginally significant ($p = 0.058$). Ranking controls have the expected signs: conditional on being a set down, stronger teams are more likely to complete the comeback.

5. Discussion and Next Steps

What the data say

- Language proximity is positive and statistically significant for match wins in the updated filtered sample, but not robustly positive for the pressure-outcome models.
- Same nationality is actually slightly negative for match win in the descriptor (losers slightly more same-nationality), and the regression coefficient, while positive, is imprecisely estimated.
- Homophily shares do not rise in the run-up to the Olympics — if anything they fall in 2024–25, driven by draw expansion.

Limitations and extensions

- **Sample size:** ~1,700 Grand Slam matches. Power is limited for detecting small homophily effects (≤ 3–5 pp).
- **Language collinearity:** same language and language proximity are highly correlated; consider a single index.
- **Draw expansion 2024–25:** the larger sample changes composition and may mask cycle-level patterns; restrict to comparable round ranges for a cleaner trend.
- **Conditioning on partnership formation:** selection — same-country pairs may self-select into tournaments — is not yet addressed.
- **Pair fixed effects:** a within-pair panel (same partners over time) would better isolate performance from selection.

Generated automatically from `build_report.py`. All regressions run in Python (statsmodels). Data: ATP Tour Grand Slam doubles 2018–2025 (excl. 2020).